Nptel Online Certification Course: Computer Vision
Week 3: Single View Camera Geometry
Type of questions: MCQ/MSQ
Number of questions: 10 **Total marks: 10 X 2 = 20**

QUESTION 1: **Type:MCQ**

A camera center is at the origin and the image plane is parallel to the XY-plane at distance $f = 0.8$. The perspective projection of the world point $(2, 4, 5)$ is:

(a) $\left(\frac{8}{25}, \frac{16}{25}\right)$

(b) $\left(\frac{2}{5}, \frac{4}{5}\right)$

(c) $\left(\frac{5}{8}, \frac{5}{4}\right)$

(d) $\left(\frac{4}{5}, \frac{8}{5}\right)$

Correct Answer: (a)
Solution: Using perspective projection,

$$(x, y) = \left(\frac{fX}{Z}, \frac{fY}{Z}\right)$$

QUESTION 2: **Type:MCQ**

If the principal point is $(1, 2)$, what are the image coordinates of the world point $(4, 6, 8)$ for a pinhole camera with $f = 1$?

(a) $(1.25, 2.5)$

(b) $(2, 2.5)$

(c) $(1.5, 2.75)$

(d) $(1.75, 2.75)$

Correct Answer: (c)
Solution: With principal point (p_x, p_y),

$$x = \frac{fX}{Z} + p_x, \qquad y = \frac{fY}{Z} + p_y$$

QUESTION 3: **Type:MCQ**

Given $P = K[R \mid -RC]$ where $f = 2$, principal point $(1, 3)$, $R = I$, and camera center $C = (2, 1, 1)$, find the projection matrix.

(a) $\begin{bmatrix} 2 & 0 & 1 & -4 \\ 0 & 2 & 3 & -6 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 0 & 1 & -6 \\ 0 & 2 & 3 & -8 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 0 & 1 & -5 \\ 0 & 2 & 3 & -5 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

(d) $\begin{bmatrix} 2 & 0 & 1 & -3 \\ 0 & 2 & 3 & -5 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

Correct Answer: (c)

Solution: The calibration matrix is

$$K = \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

Since $R = I$,

$$P = K[I \mid -C]$$

and

$$-C = (-2, -1, -1)^T$$

$$K(-C) = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -5 \\ -5 \\ -1 \end{bmatrix}$$

$$P = \begin{bmatrix} 2 & 0 & 1 & -5 \\ 0 & 2 & 3 & -5 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

QUESTION 4: **Type:MCQ**

For the projection matrix

$$P = \begin{bmatrix} 1 & 0 & 2 & -4 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & -1 \end{bmatrix},$$

find the camera center in homogeneous coordinates.

(a) $(4, 2, 1, 1)$

(b) $(2, 1, 1, 1)$

(c) $(1, 1, 1, 1)$

(d) $(2, 2, 1, 1)$

Correct Answer: (b)

Solution: Let M be the left 3×3 block and p_4 be the last column. Then

$$C = (-M^{-1} p_4, 1)$$

Here,

$$M = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad p_4 = \begin{bmatrix} -4 \\ -2 \\ -1 \end{bmatrix}$$

QUESTION 5: **Type:MCQ**

If the principal point (p_x, p_y), focal length f, orientation R, and camera center C of a pinhole camera are given by $(2, 1)$, 1,

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.8 & 0.6 \\ 0 & -0.6 & 0.8 \end{bmatrix}$$

and $(1, 2, 3)$, respectively, compute the 3×4 projection matrix P.

(a)
$$\begin{bmatrix} 1 & -1.2 & 1.6 & -3.4 \\ 0 & 0.2 & 1.4 & -4.6 \\ 0 & -0.6 & 0.8 & -1.2 \end{bmatrix}$$

(b)
$$\begin{bmatrix} 1 & -1.2 & 1.6 & -2.4 \\ 0 & 0.2 & 1.4 & -3.6 \\ 0 & -0.6 & 0.8 & -1.2 \end{bmatrix}$$

(c)
$$\begin{bmatrix} 1 & -1.2 & 1.6 & -3.4 \\ 0 & 0.2 & 1.4 & -3.6 \\ 0 & -0.6 & 0.8 & -2.4 \end{bmatrix}$$

(d)
$$\begin{bmatrix} 1 & -1.2 & 1.6 & -1.4 \\ 0 & 0.2 & 1.4 & -4.6 \\ 0 & -0.6 & 0.8 & -1.2 \end{bmatrix}$$

Correct Answer: (a)
Detailed Solution: The projection matrix is

$$P = KR[I \mid -C]$$

where the calibration matrix is

$$K = \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

First compute

$$KR = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.8 & 0.6 \\ 0 & -0.6 & 0.8 \end{bmatrix} = \begin{bmatrix} 1 & -1.2 & 1.6 \\ 0 & 0.2 & 1.4 \\ 0 & -0.6 & 0.8 \end{bmatrix}.$$

Now,

$$-C = (-1, -2, -3)^T.$$

So the last column is

$$KR(-C) = \begin{bmatrix} 1 & -1.2 & 1.6 \\ 0 & 0.2 & 1.4 \\ 0 & -0.6 & 0.8 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \\ -3 \end{bmatrix} = \begin{bmatrix} -3.4 \\ -4.6 \\ -1.2 \end{bmatrix}.$$

QUESTION 6: **Type:MCQ**

Given the projection matrix

$$P = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix},$$

find the vanishing point of a line with direction ratio $2 : 3 : 4$.

(a) $(1.5, 2.5)$

(b) $(1, 2)$

(c) $(2, 3)$

(d) $(0.25, 0.75)$

Correct Answer: (a)

Solution: The vanishing point is

$$v = P(a, b, c, 0)^T$$

with $(a, b, c) = (2, 3, 4)$. Thus,

$$v = \begin{bmatrix} 2+4 \\ 6+4 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 4 \end{bmatrix}$$

In inhomogeneous coordinates,

$$\left(\frac{6}{4}, \frac{10}{4}\right) = (1.5, 2.5)$$

QUESTION 7: **Type:MCQ**

For a camera center $C = (1, 2, 3)$ and direction vector $(2, 1, 2)$, find the scene point at distance 6 units from the center along that ray.

(a) $(2, 3, 4)$

(b) $(3, 3, 5)$

(c) $(4, 3, 6)$

(d) $(5, 4, 7)$

Correct Answer: (d)

Solution: The unit direction vector is

$$\frac{(2, 1, 2)}{\sqrt{2^2 + 1^2 + 2^2}} = \frac{(2, 1, 2)}{3}$$

So the point at distance 6 is

$$X = C + 6 \cdot \frac{(2, 1, 2)}{3}$$

$$X = (1, 2, 3) + 6 \cdot \frac{(2, 1, 2)}{3} = (5, 4, 7)$$

QUESTION 8: **Type:MCQ**

Given two image points $x_1 = (1,\ 0)$, $x_2 = (2,\ 2)$ and camera center O at the origin, find the unit normal to the plane Ox_1x_2.

a) $\left(-\frac{2}{3},\ \frac{1}{3},\ \frac{2}{3}\right)$

b) $\left(\frac{2}{3},\ -\frac{1}{3},\ \frac{2}{3}\right)$

c) $\left(-\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}\right)$

d) $\left(\frac{1}{3},\ \frac{1}{3},\ \frac{1}{3}\right)$

Correct Answer: (a)

The plane Ox_1x_2 passes through the camera center $O = (0, 0, 0)$ and x_1 and x_2. In homogeneous coordinates obtain the 3D rays:

$$\mathbf{r}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad \mathbf{r}_2 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

The normal to the plane is given by the cross product:

$$\mathbf{n} = \mathbf{r}_1 \times \mathbf{r}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{vmatrix} = \mathbf{i}(0 \cdot 1 - 1 \cdot 2) - \mathbf{j}(1 \cdot 1 - 1 \cdot 2) + \mathbf{k}(1 \cdot 2 - 0 \cdot 2) = (-2,\ 1,\ 2)^T.$$

Magnitude:

$$\|\mathbf{n}\| = \sqrt{(-2)^2 + 1^2 + 2^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3.$$

The unit normal:

$$\hat{\mathbf{n}} = \frac{\mathbf{n}}{\|\mathbf{n}\|} = \frac{1}{3}\begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \left(-\frac{2}{3},\ \frac{1}{3},\ \frac{2}{3}\right).$$

QUESTION 9: **Type:MCQ**

For a camera with $f = 1$, principal point $(2, 1)$, and world point $(5, 3, 5)$, what is the image point?

(a) $(3, \frac{9}{5})$

(b) $(3, \frac{8}{5})$

(c) $(\frac{7}{5}, \frac{8}{5})$

(d) $(\frac{8}{5}, 3)$

Correct Answer: (b)

Solution: Using

$$x = \frac{fX}{Z} + p_x, \qquad y = \frac{fY}{Z} + p_y$$

QUESTION 10: **Type:MCQ**

Consider a pinhole camera with camera center at the origin and image plane $Z = f$. A 3D point $X = (2, 4, 2)$ is projected onto the image plane with $f = 1$. Which statements are correct?

(a) The image point is $(2, 4)$.

(b) The projected point lies on the ray passing through the camera center and the 3D point.

(c) If the focal length is doubled, the image point also doubles in magnitude.

(d) The projection preserves straight lines in 3D not passing through the camera center as straight lines in the image.

Correct options: (b), (c), (d)
Solution: For a pinhole camera,

$$(x, y) = \left(\frac{fX}{Z}, \frac{fY}{Z}\right).$$

With $f = 1$ and $X = (2, 4, 2)$,

$$(x, y) = \left(\frac{2}{2}, \frac{4}{2}\right) = (1, 2).$$

So option (a) is false. The projection lies on the same ray from the camera center to the point, so (b) is true. Since image coordinates are proportional to f, doubling f doubles the projected coordinates, so (c) is true. A central projection maps 3D straight lines to 2D straight lines, so (d) is true.
